

Feature Engineering

Notebook 5 set the statistical baseline: SARIMAX models the time dependence **inside** the algorithm. The next step of the benchmark is a Machine Learning model. A standard regressor has no notion of time or order: it sees a table of rows $X \rightarrow y$, so the temporal structure has to be written into the columns of X .

This notebook builds that table for the **same series as notebook 5**: monthly international arrivals to Brazil, aggregated over all access routes, origins and destination states. It covers feature engineering only; model training comes next.

Each step follows the same layout:

1. **What** is built and **why** it is relevant for this series.
2. **How**: the book helper used and the parameters chosen.
3. **Output**: the new columns, a sample of their values and a plot or check.

Setup

```
In [1]: import json
import os
import sys

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns
from IPython.display import display
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
from statsmodels.tsa.stattools import acf, pacf

ROOT_PATH = os.path.dirname(os.getcwd())
DATA_FOLDER_PATH = os.path.join(ROOT_PATH, 'data', 'processed')

COVID_START = pd.Timestamp('2020-01-01')
COVID_END = pd.Timestamp('2021-12-31')
POST_COVID_START = pd.Timestamp('2022-01-01')
COVID_RECOVERY_END = pd.Timestamp('2022-06-30')

TARGET = 'arrival_count'
TS_ID = 'series_id'
FREQ = 'ME'
SEASONAL_PERIOD = 12

sns.set_palette('colorblind')
pd.set_option('display.max_columns', 40)
```

Book Helpers

The feature functions come from the companion repository of:

Joseph, M. & Tackes, J. *Modern Time Series Forecasting with Python*, 2nd ed.
(Packt, 2024) — [PacktPublishing/Modern-Time-Series-Forecasting-with-Python-2E](#), MIT License

Helper	Module
add_lags	src/feature_engineering/autoregressive_features.py
add_rolling_features	src/feature_engineering/autoregressive_features.py
add_seasonal_rolling_features	src/feature_engineering/autoregressive_features.py
add_ewma	src/feature_engineering/autoregressive_features.py
add_temporal_features	src/feature_engineering/temporal_features.py
add_fourier_features	src/feature_engineering/temporal_features.py

Notes on using them here:

- **ts_id is always passed**, even though there is a single series (a constant `series_id` column). Without `ts_id`, `add_seasonal_rolling_features` calls `Series.transform(lambda ...)`, which pandas 2.x applies element by element (`ValueError: Function did not transform`). The grouped path passes the whole series.
- **add_ewma is called with spans**. With only `alphas`, the helper reads an undefined variable (`use_spans`) and fails.
- **add_fourier_features is called with max_value=12**. Without it, the helper raises instead of warning.
- `add_seasonal_rolling_features` needs the `window-ops` package (a project dependency).

The repository is expected as a sibling folder of this project (override with the `MTSF_REPO_PATH` environment variable).

```
In [2]: # Helpers credited to: Joseph, M. & Tackes, J. "Modern Time Series Forecasting with
# Python", 2nd ed. (Packt, 2024) – MIT License
```

```
# github.com/PacktPublishing/Modern-Time-Series-Forecasting-with-Python-2E
MTSF_REPO_PATH = os.getenv(
    'MTSF_REPO_PATH',
    os.path.join(
        os.path.dirname(ROOT_PATH), 'Modern-Time-Series-Forecasting-with-Python-2E'
    ),
)
if MTSF_REPO_PATH not in sys.path:
    sys.path.insert(0, MTSF_REPO_PATH)

from src.feature_engineering.autoregressive_features import ( # noqa: E402
    add_ewma,
    add_lags,
    add_rolling_features,
    add_seasonal_rolling_features,
)
from src.feature_engineering.temporal_features import ( # noqa: E402
    add_fourier_features,
    add_temporal_features,
)
```

Step 1 — Casting the Series as a Supervised Regression

What. The series $\{y_1, \dots, y_T\}$ becomes a table: each row is a month t , the target column is y_t , and the feature columns summarize what was known **before** t .

How.

- Monthly totals are built from `train.parquet` and `val.parquet`, the same aggregation as notebook 5. The validation set is the 12 months before the test year (notebook 2), and the dates used below (`TRAIN_END`, `VAL_START`, `HORIZON`) are read from these files, so the notebook follows the data when it is updated. The **test set is never read here**: it is kept for the final evaluation only.
- Train and validation are **joined** before building features, and a `split` column keeps them apart. Features for the first validation month need the values of the last training months, which only exist in the training file. If the validation set were processed alone, its first rows would have no lags.
- The joined table is safe only if no feature of row t reads y_t or a later value. Every target-derived feature is shifted for that reason, and Step 11 tests it.

```
In [3]: def load_monthly(file_name, split):
    raw = pd.read_parquet(os.path.join(DATA_FOLDER_PATH, file_name))
    monthly = raw.set_index('date')[[TARGET]].resample(FREQ).sum()
    monthly['split'] = split
    return monthly

monthly = pd.concat([
    load_monthly('train.parquet', 'train'),
```

```

load_monthly('val.parquet', 'val'),
]).reset_index()
monthly[TS_ID] = 'brazil_total'
monthly = monthly[['date', TS_ID, 'split', TARGET]]

# every date of this notebook is derived from the split (notebook 2)
TRAIN_END = monthly.loc[monthly['split'] == 'train', 'date'].max()
VAL_START = monthly.loc[monthly['split'] == 'val', 'date'].min()
HORIZON = int((monthly['split'] == 'val').sum())
assert VAL_START == TRAIN_END + pd.offsets.MonthEnd(1), 'validation must follow tra

for split, part in monthly.groupby('split', sort=False):
    print(
        f'{split:<5}: {part["date"].min():%b %Y} → {part["date"].max():%b %Y} '
        f'({len(part)} months)'
    )
print(f'HORIZON = {HORIZON} months')
display(monthly.tail(15))

```

train: Jan 1989 → Aug 2024 (428 months)
val : Sep 2024 → Aug 2025 (12 months)
HORIZON = 12 months

	date	series_id	split	arrival_count
425	2024-06-30	brazil_total	train	332474.0
426	2024-07-31	brazil_total	train	437103.0
427	2024-08-31	brazil_total	train	417940.0
428	2024-09-30	brazil_total	val	445389.0
429	2024-10-31	brazil_total	val	508738.0
430	2024-11-30	brazil_total	val	560732.0
431	2024-12-31	brazil_total	val	806478.0
432	2025-01-31	brazil_total	val	1483669.0
433	2025-02-28	brazil_total	val	1326884.0
434	2025-03-31	brazil_total	val	929096.0
435	2025-04-30	brazil_total	val	686239.0
436	2025-05-31	brazil_total	val	461341.0
437	2025-06-30	brazil_total	val	444882.0
438	2025-07-31	brazil_total	val	620143.0
439	2025-08-31	brazil_total	val	576049.0

In [4]:

```

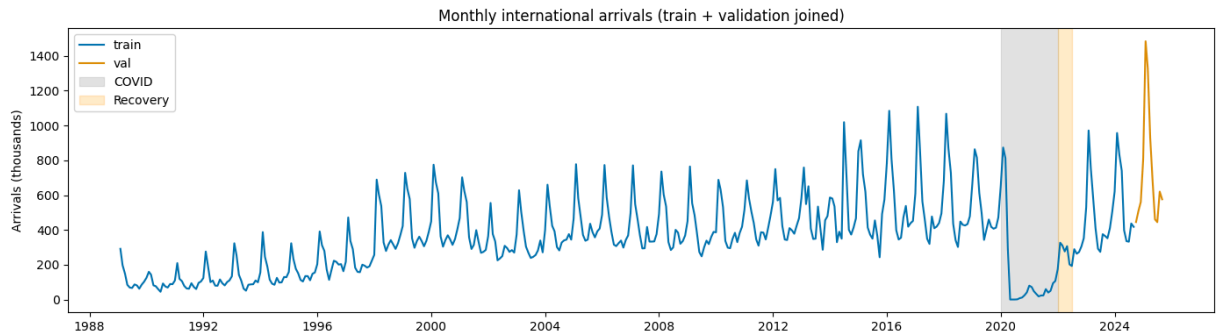
fig, ax = plt.subplots(figsize=(14, 4))
for split, part in monthly.groupby('split', sort=False):
    ax.plot(part['date'], part[TARGET] / 1e3, label=split)
ax.axvspan(COVID_START, COVID_END, color='grey', alpha=0.2, label='COVID')

```

```

ax.axvspan(
    POST_COVID_START, COVID_RECOVERY_END, color='orange', alpha=0.2, label='Recover
')
ax.set_title('Monthly international arrivals (train + validation joined)')
ax.set_ylabel('Arrivals (thousands)')
ax.legend()
plt.tight_layout()
plt.show()

```



Sliding window illustration. With a memory of $M = 3$ months, the rows of the table look like the frame below: the three previous values are the inputs and the current value is the target. The following steps generalize this idea to lags, windows and summaries chosen for this series.

```

In [5]: M = 3
window_example = pd.DataFrame(
    {f'y(t-{k})': monthly[TARGET].shift(k) for k in range(M, 0, -1)}
    | {'y(t) → target': monthly[TARGET]})
).set_index(monthly['date'].dt.strftime('%b %Y'))
window_example.iloc[M : M + 6]

```

```

Out[5]:

```

	y(t-3)	y(t-2)	y(t-1)	y(t) → target
Apr 1989	291566.0	197348.0	149001.0	84867.0
May 1989	197348.0	149001.0	84867.0	69293.0
Jun 1989	149001.0	84867.0	69293.0	66838.0
Jul 1989	84867.0	69293.0	66838.0	87054.0
Aug 1989	69293.0	66838.0	87054.0	81535.0
Sep 1989	66838.0	87054.0	81535.0	62629.0

	date				
	Apr 1989	291566.0	197348.0	149001.0	84867.0
	May 1989	197348.0	149001.0	84867.0	69293.0
	Jun 1989	149001.0	84867.0	69293.0	66838.0
	Jul 1989	84867.0	69293.0	66838.0	87054.0
	Aug 1989	69293.0	66838.0	87054.0	81535.0
	Sep 1989	66838.0	87054.0	81535.0	62629.0

Step 2 — Choosing the Memory: ACF and PACF

What. The lags to include depend on how far back the series "remembers".

- **ACF:** the correlation between y_t and y_{t-k} .

- **PACF**: the same correlation after removing the effect of the intermediate lags. It points to the lags that add **new** information.

How. Both are computed on the **pre-COVID training period** (1989–2019). The 2020–2021 collapse would otherwise dominate the correlations. The significant lags are those outside the 95% confidence band.

```
In [6]: pre_covid = monthly.loc[
    (monthly['split'] == 'train') & (monthly['date'] < COVID_START), TARGET
]
N_LAGS = 36

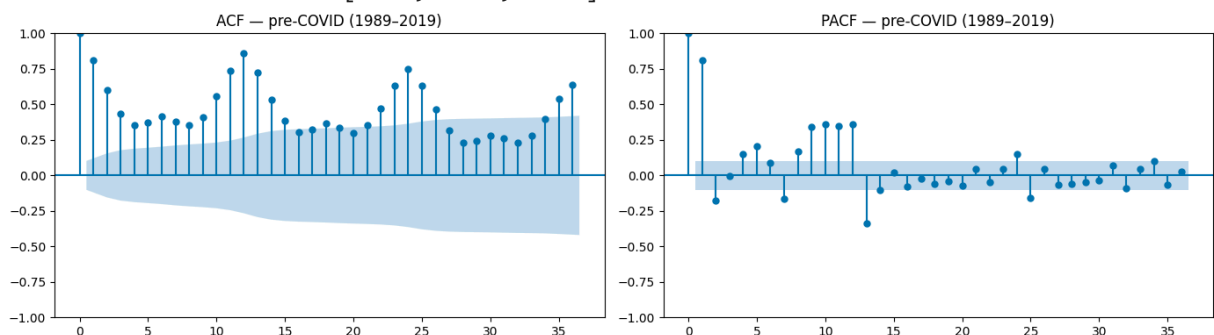
def significant_lags(values, conf_int):
    half_width = conf_int[:, 1] - values
    return {
        lag: round(float(values[lag]), 2)
        for lag in range(1, len(values))
        if abs(values[lag]) > half_width[lag]
    }

acf_values, acf_ci = acf(pre_covid, nlags=N_LAGS, alpha=0.05)
pacf_values, pacf_ci = pacf(pre_covid, nlags=N_LAGS, alpha=0.05)
print('Significant PACF lags:', significant_lags(pacf_values, pacf_ci))
print('ACF at 12 / 24 / 36 :', [round(float(acf_values[k]), 2) for k in (12, 24, 36)])

fig, axes = plt.subplots(1, 2, figsize=(14, 4))
plot_acf(pre_covid, lags=N_LAGS, ax=axes[0], title='ACF — pre-COVID (1989–2019)')
plot_pacf(pre_covid, lags=N_LAGS, ax=axes[1], title='PACF — pre-COVID (1989–2019)')
plt.tight_layout()
plt.show()
```

Significant PACF lags: {1: 0.82, 2: -0.18, 4: 0.15, 5: 0.21, 7: -0.17, 8: 0.17, 9: 0.36, 10: 0.38, 11: 0.38, 12: 0.43, 13: -0.33, 14: -0.12, 24: 0.18, 25: -0.17, 32: -0.11, 34: 0.13}

ACF at 12 / 24 / 36 : [0.86, 0.75, 0.64]



Reading.

- **ACF**. The ACF peaks at 12, 24 and 36, the yearly seasonality found in notebook 3. Its slow decay reflects the trend.
- **PACF lag 1** (≈ 0.82) carries the short-term momentum.

- **PACF lags 12 and 13** (the second one negative) are the typical *seasonal* + *seasonal-lag* pair of monthly data. Lag 24 still adds information.
- **PACF lags 9–11** are also significant. They sit on the approach to the seasonal peak and are mostly summarized by the rolling and seasonal features built later.

Step 3 — Forecasting Strategies and the Leakage Rule

What. The forecast horizon is `HORIZON` = 12 months (the validation year). When forecasting the last validation month from the last training month, the values of the 11 months in between are **not known**. That fixes which lags are legal:

Strategy	Minimum shift	How the next notebook forecasts
recursive	1 month	One model predicts <code>t+1</code> ; the prediction is fed back as <code>lag_1</code> for <code>t+2</code> , and so on for 12 steps. The target-derived features of the validation rows must be rebuilt from predictions.
direct	12 months (= <code>HORIZON</code>)	Every feature is already known for the whole horizon, so the 12 validation months are predicted in one pass, without feeding predictions back.

How. One configuration per strategy. Every target-derived feature is shifted by at least `min_shift` .

- **Lags** are chosen from the PACF of Step 2: 1, 2, 12, 13, 24 and 36 for recursive, and only those ≥ 12 for direct.
- **Rolling windows** are shifted by `min_shift` before the window is applied.
- The other parameters are shared by both strategies.

The two feature tables are built side by side in each of the following steps.

```
In [7]: BASE_LAGS = [1, 2, 12, 13, 24, 36]

STRATEGIES = {
    'recursive': {'min_shift': 1},
    'direct': {'min_shift': HORIZON},
}
for cfg in STRATEGIES.values():
    cfg['lags'] = [lag for lag in BASE_LAGS if lag >= cfg['min_shift']]

ROLLING_WINDOWS = [3, 6, 12]
ROLLING_AGGS = ['mean', 'std', 'min', 'max']
SEASONAL_WINDOWS = [2, 3] # the same month of the 2 / 3 previous years
SEASONAL_AGGS = ['mean', 'std']
EWMA_SPANS = [3, 6, 12]
FOURIER_TERMS = 3

display(pd.DataFrame(STRATEGIES).T)
```

```

# one working table per strategy, filled step by step
features = {name: monthly.copy() for name in STRATEGIES}
feature_groups = {name: {} for name in STRATEGIES}

# around the train/validation boundary: the last 3 train and first 3 validation mon
BOUNDARY = slice(TRAIN_END - pd.offsets.MonthEnd(2), VAL_START + pd.offsets.MonthEn

def show_step(group, columns=None, rows=BOUNDARY):
    """Summarize a step's new features and show them around the train/val boundary.
    display(
        pd.DataFrame({
            name: {
                'n_features': len(groups[group]),
                'features': ', '.join(groups[group]),
            }
            for name, groups in feature_groups.items()
        }).T
    )
    for name, frame in features.items():
        cols = columns or feature_groups[name][group]
        cols = [c for c in cols if c in frame.columns]
        print(f'{name} - sample around the train/validation boundary')
        display(frame.set_index('date').loc[rows, ['split', TARGET, *cols]])

```

	min_shift	lags
recursive	1	[1, 2, 12, 13, 24, 36]
direct	12	[12, 13, 24, 36]

Step 4 — Lags (Time Delay Embedding)

What. A lag y_{t-k} places the value observed k months earlier on the row of month t . It encodes the autoregressive structure:

- `lag_1`: momentum, i.e. where the series was last month.
- `lag_12`: the same month last year, which carries the seasonal level.
- `lag_13`, `lag_24`, `lag_36`: the seasonal pattern of previous years.

How. `add_lags(df, lags, column, ts_id)` (book) shifts the series by each lag.

Output. Near the train/validation boundary, the recursive `lag_1` of the second validation month holds the **actual** value of the first validation month. That value is legitimate only in one-step-ahead mode, which is why Step 13 masks it for the recursive validation rows. For the direct strategy, every lag of a validation row points to the training period.

```

In [8]: def add_lag_features(df, cfg):
        return add_lags(df, lags=cfg['lags'], column=TARGET, ts_id=TS_ID)

```

```
for name, cfg in STRATEGIES.items():
    features[name], feature_groups[name]['lags'] = add_lag_features(features[name],
    show_step('lags')
```

	n_features	features
recursive	6	arrival_count_lag_1, arrival_count_lag_2, arri...
direct	4	arrival_count_lag_12, arrival_count_lag_13, ar...

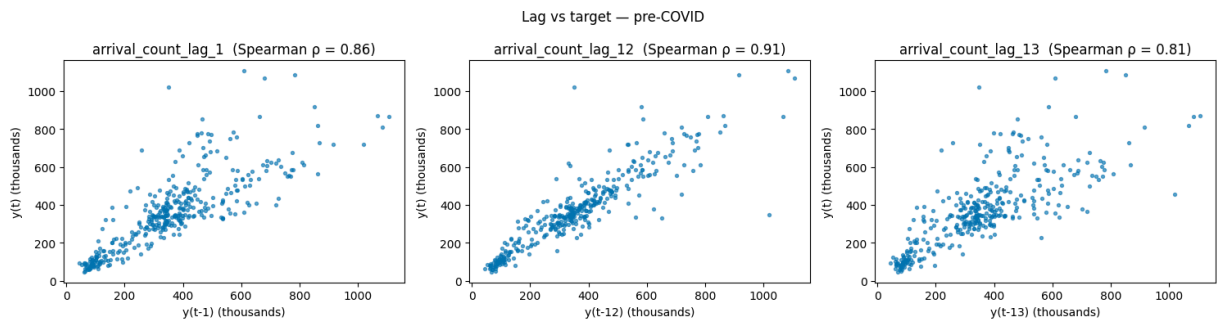
recursive – sample around the train/validation boundary

	split	arrival_count	arrival_count_lag_1	arrival_count_lag_2	arrival_count_lag_12	arrival_count_lag_13
date						
2024-06-30	train	332474.0	335652.0	398587.0	274302.0	292375.0
2024-07-31	train	437103.0	332474.0	335652.0	375648.0	274302.0
2024-08-31	train	417940.0	437103.0	332474.0	365041.0	375648.0
2024-09-30	val	445389.0	417940.0	437103.0	352355.0	365041.0
2024-10-31	val	508738.0	445389.0	417940.0	409924.0	352355.0
2024-11-30	val	560732.0	508738.0	445389.0	504395.0	409924.0

direct – sample around the train/validation boundary

	split	arrival_count	arrival_count_lag_12	arrival_count_lag_13	arrival_count_lag_24	arrival_count_lag_25
date						
2024-06-30	train	332474.0	274302.0	292375.0	192909.0	289581.0
2024-07-31	train	437103.0	375648.0	274302.0	289581.0	263626.0
2024-08-31	train	417940.0	365041.0	375648.0	263626.0	273004.0
2024-09-30	val	445389.0	352355.0	365041.0	273004.0	305644.0
2024-10-31	val	508738.0	409924.0	352355.0	305644.0	352458.0
2024-11-30	val	560732.0	504395.0	409924.0	352458.0	

```
In [9]: lag_view = features['recursive'].query('date < @COVID_START')
fig, axes = plt.subplots(1, 3, figsize=(15, 4))
for ax, lag in zip(axes, [1, 12, 13]):
    col = f'{TARGET}_lag_{lag}'
    rho = lag_view[[col, TARGET]].corr(method='spearman').iloc[0, 1]
    ax.scatter(lag_view[col] / 1e3, lag_view[TARGET] / 1e3, s=8, alpha=0.6)
    ax.set_title(f'{col} (Spearman  $\rho$  = {rho:.2f})')
    ax.set_xlabel(f'y(t-{lag}) (thousands)')
    ax.set_ylabel('y(t) (thousands)')
fig.suptitle('Lag vs target — pre-COVID')
plt.tight_layout()
plt.show()
```



Step 5 — Rolling Window Aggregations

What. A single lag is noisy. A rolling window summarizes the last W observations $\{y_{\{t-W\}}, \dots, y_{\{t-1\}}\}$:

- **mean:** the local level. The 12-month mean removes the seasonality, so it tracks the trend.
- **std:** the local volatility. It is high around the COVID collapse and in high seasons.
- **min / max:** the range of the window.

How. `add_rolling_features(df, rolls, column, agg_funcs, ts_id, n_shift)` (book). The series is first shifted by `n_shift = min_shift` and then rolled. For the direct strategy, the 12-month mean of the last validation month therefore covers the last 12 training months.

Output. The plot shows the same 12-month mean for both strategies. The direct version is the recursive one moved 12 months later, which is the cost of knowing it for the whole horizon.

```
In [10]: def add_rolling_window_features(df, cfg):
    return add_rolling_features(
        df,
        rolls=ROLLING_WINDOWS,
        column=TARGET,
        agg_funcs=ROLLING_AGGS,
        ts_id=TS_ID,
        n_shift=cfg['min_shift'],
```

```

    )

for name, cfg in STRATEGIES.items():
    features[name], feature_groups[name]['rolling'] = add_rolling_window_features(
        features[name], cfg
    )

show_step(
    'rolling',
    columns=[
        f'{TARGET}_rolling_3_mean',
        f'{TARGET}_rolling_12_mean',
        f'{TARGET}_rolling_12_std',
    ],
)

```

n_features		features		
recursive	12	arrival_count_rolling_3_mean, arrival_count_ro...		
direct	12	arrival_count_rolling_3_mean, arrival_count_ro...		

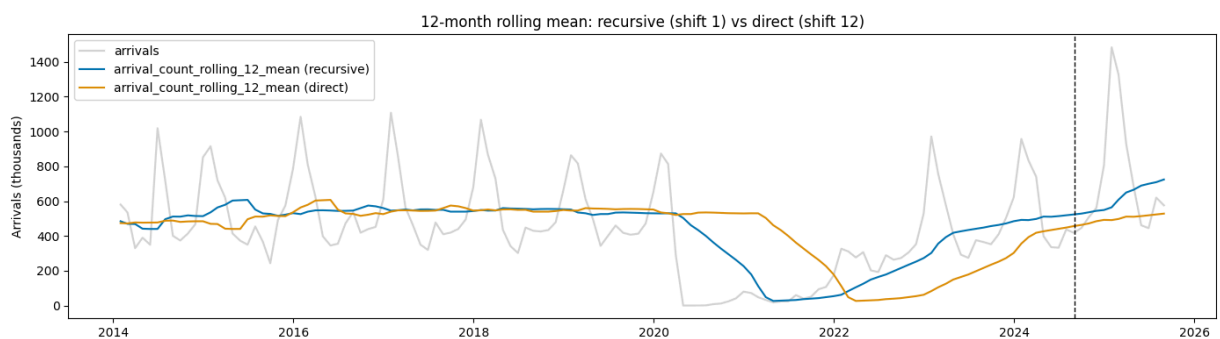
recursive – sample around the train/validation boundary

	split	arrival_count	arrival_count_rolling_3_mean	arrival_count_rolling_12_mean	arrival
date					
2024-06-30	train	332474.0	491574.000000	513966.750000	
2024-07-31	train	437103.0	355571.000000	518814.416667	
2024-08-31	train	417940.0	368409.666667	523935.666667	
2024-09-30	val	445389.0	395839.000000	528343.916667	
2024-10-31	val	508738.0	433477.333333	536096.750000	
2024-11-30	val	560732.0	457355.666667	544331.250000	

direct – sample around the train/validation boundary

	split	arrival_count	arrival_count_rolling_3_mean	arrival_count_rolling_12_mean	arrival
date					
2024-06-30	train	332474.0	325158.333333	441096.500000	
2024-07-31	train	437103.0	314108.333333	448268.750000	
2024-08-31	train	417940.0	338330.333333	456720.000000	
2024-09-30	val	445389.0	364348.000000	463332.583333	
2024-10-31	val	508738.0	375773.333333	472022.583333	
2024-11-30	val	560732.0	422224.666667	484684.000000	

```
In [11]: col = f'{TARGET}_rolling_12_mean'
fig, ax = plt.subplots(figsize=(14, 4))
view = monthly.query('date >= "2014-01-01"')
ax.plot(view['date'], view[TARGET] / 1e3, color='lightgrey', label='arrivals')
for name, frame in features.items():
    frame_view = frame.query('date >= "2014-01-01"')
    ax.plot(frame_view['date'], frame_view[col] / 1e3, label=f'{col} ({name})')
ax.axvline(TRAIN_END + pd.Timedelta(days=1), color='black', linestyle='--', linewidth=2)
ax.set_title('12-month rolling mean: recursive (shift 1) vs direct (shift 12)')
ax.set_ylabel('Arrivals (thousands)')
ax.legend()
plt.tight_layout()
plt.show()
```



Step 6 — Seasonal Rolling Aggregations

What. A seasonal rolling window skips $M = 12$ months between its elements. For month t it summarizes $\{y_{t-12}, y_{t-24}, \dots\}$: the **same calendar month** of previous years.

- **mean:** a "typical January" computed from the last 2 or 3 Januaries. This is the strongest single signal for a strongly seasonal monthly series.
- **std:** how stable that month has been across years.

How. `add_seasonal_rolling_features(df, seasonal_periods=[12], rolls, column, agg_funcs, ts_id, n_shift=1)` (book, backed by `window_ops`). Here `n_shift` counts **seasons**: the series is shifted by `1 × 12` months before the window, so the most recent element is `y_{t-12}`. This is legal for both strategies, so both use the same parameters.

Output. A manual check confirms the definition. The plot compares each January with the mean of the previous Januaries.

```
In [12]: def add_seasonal_window_features(df, cfg):
# n_shift counts seasons: shift 1 × 12 months, legal for both strategies
return add_seasonal_rolling_features(
    df,
    seasonal_periods=[SEASONAL_PERIOD],
    rolls=SEASONAL_WINDOWS,
    column=TARGET,
    agg_funcs=SEASONAL_AGGS,
    ts_id=TS_ID,
    n_shift=1,
)

for name, cfg in STRATEGIES.items():
    features[name], feature_groups[name]['seasonal_rolling'] = (
        add_seasonal_window_features(features[name], cfg)
    )

show_step('seasonal_rolling')

# manual check: the 3-year seasonal mean of the first validation month = mean of th
# same month in the 3 previous years
by_date = features['recursive'].set_index('date')
previous_years = [VAL_START - pd.offsets.MonthEnd(12 * k) for k in (1, 2, 3)]
expected = by_date.loc[previous_years, TARGET].mean()
computed = by_date.loc[VAL_START, f'{TARGET}_12_seasonal_rolling_3_mean']
print(f'{VAL_START:%b %Y} - expected {expected:,.0f} | computed {computed:,.0f}')
assert np.isclose(expected, computed)
```

	n_features	features
recursive	4	arrival_count_12_seasonal_rolling_2_mean, arri...
direct	4	arrival_count_12_seasonal_rolling_2_mean, arri...

recursive – sample around the train/validation boundary

	split	arrival_count	arrival_count_12_seasonal_rolling_2_mean	arrival_count_12_seasonal
date				
2024-06-30	train	332474.0		233605.5
2024-07-31	train	437103.0		332614.5
2024-08-31	train	417940.0		314333.5
2024-09-30	val	445389.0		312679.5
2024-10-31	val	508738.0		357784.0
2024-11-30	val	560732.0		428426.5

direct – sample around the train/validation boundary

	split	arrival_count	arrival_count_12_seasonal_rolling_2_mean	arrival_count_12_seasonal
date				
2024-06-30	train	332474.0		233605.5
2024-07-31	train	437103.0		332614.5
2024-08-31	train	417940.0		314333.5
2024-09-30	val	445389.0		312679.5
2024-10-31	val	508738.0		357784.0
2024-11-30	val	560732.0		428426.5

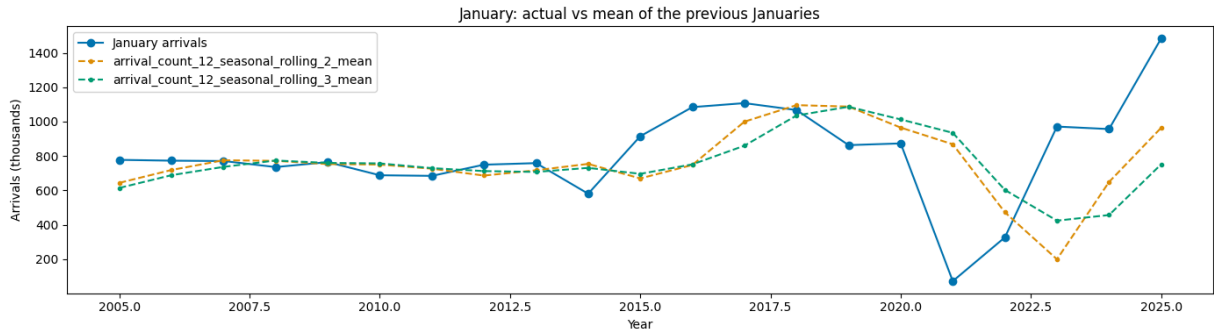
Sep 2024 – expected 225,345 | computed 225,345

```
In [13]: january = features['recursive'].query('date.dt.month == 1 and date >= "2005-01-01"')
fig, ax = plt.subplots(figsize=(14, 4))
ax.plot(
    january['date'].dt.year, january[TARGET] / 1e3, marker='o', label='January arri
)
for window in SEASONAL_WINDOWS:
    col = f'{TARGET}_12_seasonal_rolling_{window}_mean'
    ax.plot(
        january['date'].dt.year,
        january[col] / 1e3,
        marker='.',
```

```

        linestyle='--',
        label=col,
    )
ax.set_title('January: actual vs mean of the previous Januaries')
ax.set_xlabel('Year')
ax.set_ylabel('Arrivals (thousands)')
ax.legend()
plt.tight_layout()
plt.show()

```



Step 7 — Exponentially Weighted Moving Average (EWMA)

What. An EWMA averages **all** of the history, with weights that decay exponentially: $EWMA_t = \alpha \cdot y_t + (1 - \alpha) \cdot EWMA_{t-1}$. It gives long memory in one column while favouring recent months.

How. `add_ewma(df, column, spans, ts_id, n_shift)` (book), with `n_shift = min_shift` as in the rolling windows. A span `S` corresponds to $\alpha = 2 / (1 + S)$:

span	α	half-life (months)
3	0.50	≈ 1.0
6	0.29	≈ 2.1
12	0.15	≈ 4.3

Output. The shorter span follows the seasonal swings. The longer spans smooth them toward the level.

```

In [14]: def add_ewma_features(df, cfg):
            # spans must be passed: the helper fails when only alphas are given
            return add_ewma(
                df,
                column=TARGET,
                alphas=None,
                spans=EWMA_SPANS,
                ts_id=TS_ID,
                n_shift=cfg['min_shift'],
            )

```

```
for name, cfg in STRATEGIES.items():
    features[name], feature_groups[name]['ewma'] = add_ewma_features(
        features[name], cfg
    )

show_step('ewma')
```

	n_features	features
recursive	3	arrival_count_ewma_span_3, arrival_count_ewma_...
direct	3	arrival_count_ewma_span_3, arrival_count_ewma_...

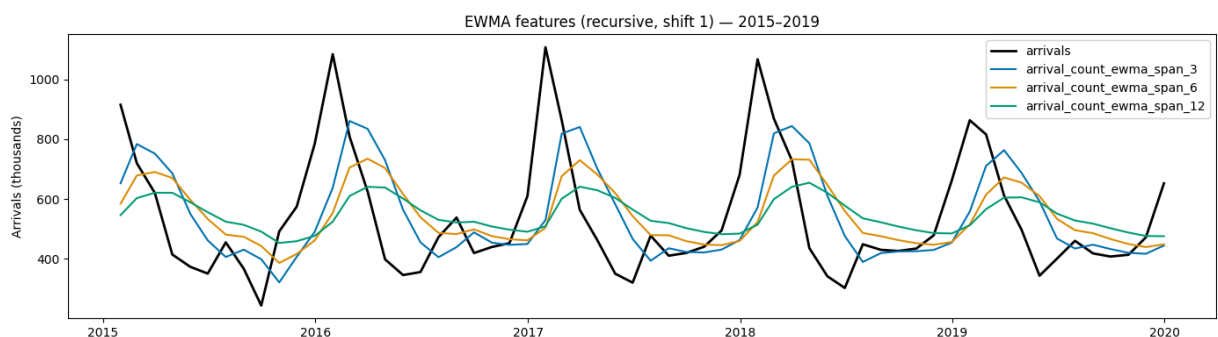
recursive – sample around the train/validation boundary

	split	arrival_count	arrival_count_ewma_span_3	arrival_count_ewma_span_6	arrival_cou
date					
2024-06-30	train	332474.0	458655.934161	532149.381254	
2024-07-31	train	437103.0	395564.967080	475099.272324	
2024-08-31	train	417940.0	416333.983540	464243.194517	
2024-09-30	val	445389.0	417136.991770	451013.710369	
2024-10-31	val	508738.0	431262.995885	449406.650264	
2024-11-30	val	560732.0	470000.497943	466358.464474	

direct – sample around the train/validation boundary

	split	arrival_count	arrival_count_ewma_span_3	arrival_count_ewma_span_6	arrival_count_ewma_span_12
2024-06-30	train	332474.0	342843.160791	413463.392736	
2024-07-31	train	437103.0	359245.580396	402658.994811	
2024-08-31	train	417940.0	362143.290198	391910.996294	
2024-09-30	val	445389.0	357249.145099	380609.283067	
2024-10-31	val	508738.0	383586.572549	388984.916476	
2024-11-30	val	560732.0	443990.786275	421959.226055	

```
In [15]: view = features['recursive'].query('"2015-01-01" <= date < "2020-01-01"')
fig, ax = plt.subplots(figsize=(14, 4))
ax.plot(view['date'], view[TARGET] / 1e3, color='black', linewidth=2, label='arrivals')
for span in EWMA_SPANS:
    col = f'{TARGET}_ewma_span_{span}'
    ax.plot(view['date'], view[col] / 1e3, label=col)
ax.set_title('EWMA features (recursive, shift 1) - 2015-2019')
ax.set_ylabel('Arrivals (thousands)')
ax.legend()
plt.tight_layout()
plt.show()
```



Step 8 — Calendar Features and Elapsed Time (Temporal Embedding)

What. Temporal embedding builds features from the **timestamp**, not from past target values, so they are known for any future date.

- **Month and quarter** capture the tourism calendar: the southern-hemisphere summer and Carnival (January–March) and the July holidays.

- **Elapsed** is a monotonically increasing time counter, in seconds since the epoch. It lets linear and neural models estimate a long-term trend. Trees **cannot extrapolate** it beyond the training range, which is why the target transformation of notebook 7 matters for XGBoost.

How. `add_temporal_features(df, field_name='date', frequency='ME', add_elapsed=True, prefix='cal', drop=False)` (book). For a month-end frequency, the helper returns month, quarter, the quarter/year start and end flags, and elapsed time.

- The dates are **month ends**, so `Is_quarter_start` and `Is_year_start` are always `False`. These constant columns carry no information and are dropped.
- Boolean flags are cast to `0/1`.
- `Elapsed` comes back as `object` dtype and is cast to `int64`.

Output. The box plot shows why the month matters: the level of arrivals depends strongly on the calendar month.

```
In [16]: # month-end dates never start a quarter or a year → the helper's flags are constant
CONSTANT_ON_MONTH_END = ['cal_Is_quarter_start', 'cal_Is_year_start']

def add_calendar_features(df, cfg):
    df, added = add_temporal_features(
        df,
        field_name='date',
        frequency=FREQ,
        add_elapsed=True,
        prefix='cal',
        drop=False,
    )
    assert all((df[c] == df[c].iloc[0]).all() for c in CONSTANT_ON_MONTH_END)
    df = df.drop(columns=CONSTANT_ON_MONTH_END)
    added = [c for c in added if c not in CONSTANT_ON_MONTH_END]
    flag_cols = [c for c in added if c.startswith('cal_Is_')]
    df[flag_cols] = df[flag_cols].astype(int)
    df['cal_Elapsed'] = df['cal_Elapsed'].astype('int64')
    return df, added

for name, cfg in STRATEGIES.items():
    features[name], feature_groups[name]['calendar'] = add_calendar_features(
        features[name].copy(), cfg
    )

show_step('calendar')
```

	n_features	features
recursive	5	cal_Month, cal_Quarter, cal_Is_quarter_end, ca...
direct	5	cal_Month, cal_Quarter, cal_Is_quarter_end, ca...

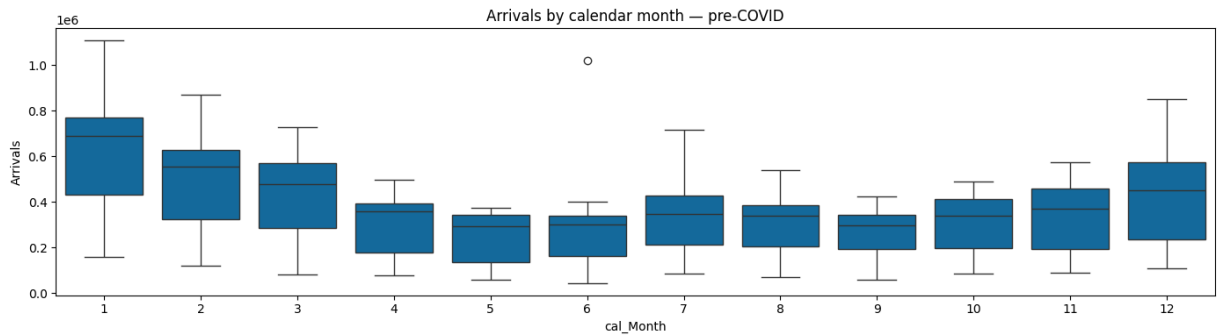
recursive – sample around the train/validation boundary

	split	arrival_count	cal_Month	cal_Quarter	cal_Is_quarter_end	cal_Is_year_end	cal_El
date							
2024-06-30	train	332474.0	6	2	1	0	17197
2024-07-31	train	437103.0	7	3	0	0	17223
2024-08-31	train	417940.0	8	3	0	0	17250
2024-09-30	val	445389.0	9	3	1	0	17276
2024-10-31	val	508738.0	10	4	0	0	17303
2024-11-30	val	560732.0	11	4	0	0	17329

direct – sample around the train/validation boundary

	split	arrival_count	cal_Month	cal_Quarter	cal_Is_quarter_end	cal_Is_year_end	cal_El
date							
2024-06-30	train	332474.0	6	2	1	0	17197
2024-07-31	train	437103.0	7	3	0	0	17223
2024-08-31	train	417940.0	8	3	0	0	17250
2024-09-30	val	445389.0	9	3	1	0	17276
2024-10-31	val	508738.0	10	4	0	0	17303
2024-11-30	val	560732.0	11	4	0	0	17329

```
In [17]: fig, ax = plt.subplots(figsize=(14, 4))
sns.boxplot(
    data=features['recursive'].query('date < @COVID_START'),
    x='cal_Month',
    y=TARGET,
    ax=ax,
)
ax.set_title('Arrivals by calendar month – pre-COVID')
ax.set_ylabel('Arrivals')
plt.tight_layout()
plt.show()
```



Step 9 — Fourier Terms (Trigonometric Seasonality)

What. The month is also encoded as sine/cosine pairs: $\sin(2\pi \cdot n \cdot \text{month} / 12)$ and $\cos(2\pi \cdot n \cdot \text{month} / 12)$ for $n = 1, \dots, N$.

- Unlike the integer month, December (12) and January (1) end up **close** to each other.
- The curves are smooth, so a model can build the seasonal shape from a few columns.
- More terms (n) capture sharper seasonal patterns. $N = 3$ covers yearly, half-yearly and quarterly cycles.

How. `add_fourier_features(df, column_to_encode='cal_Month', max_value=12, n_fourier_terms=3)` (book).

Output. The six curves over the 12 months. As noted in the study notes, whether the **calendar** or the **Fourier** encoding works better is an empirical question. Both are kept, and notebook 7 compares them (see `feature_groups`).

```
In [18]: def add_fourier_month_features(df, cfg):
# max_value must be passed: the helper raises when it has to infer it
return add_fourier_features(
    df, column_to_encode='cal_Month', max_value=12, n_fourier_terms=FOURIER_TER
)

for name, cfg in STRATEGIES.items():
    features[name], feature_groups[name]['fourier'] = add_fourier_month_features(
        features[name].copy(), cfg
    )

# the last 12 training months: one full cycle of the month
show_step('fourier', rows=slice(TRAIN_END - pd.offsets.MonthEnd(11), TRAIN_END))
```

	n_features	features
recursive	6	cal_Month_sin_1, cal_Month_sin_2, cal_Month_si...
direct	6	cal_Month_sin_1, cal_Month_sin_2, cal_Month_si...

recursive – sample around the train/validation boundary

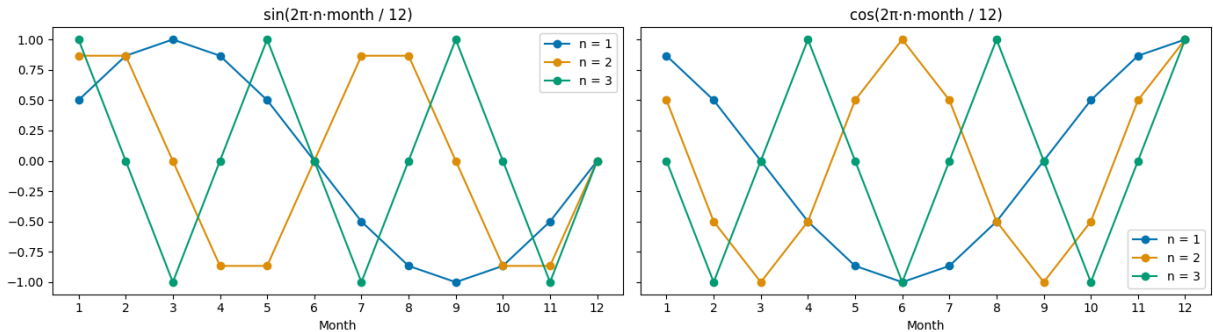
	split	arrival_count	cal_Month_sin_1	cal_Month_sin_2	cal_Month_sin_3	cal_Month_cos
date						
2023-09-30	train	352355.0	-1.000000e+00	3.673940e-16	1.000000e+00	-1.836970e-
2023-10-31	train	409924.0	-8.660254e-01	-8.660254e-01	2.388680e-15	5.000000e-
2023-11-30	train	504395.0	-5.000000e-01	-8.660254e-01	-1.000000e+00	8.660254e-
2023-12-31	train	621171.0	-2.449294e-16	-4.898587e-16	-7.347881e-16	1.000000e+
2024-01-31	train	956737.0	5.000000e-01	8.660254e-01	1.000000e+00	8.660254e-
2024-02-29	train	833306.0	8.660254e-01	8.660254e-01	1.224647e-16	5.000000e-
2024-03-31	train	740483.0	1.000000e+00	1.224647e-16	-1.000000e+00	6.123234e-
2024-04-30	train	398587.0	8.660254e-01	-8.660254e-01	-2.449294e-16	-5.000000e-
2024-05-31	train	335652.0	5.000000e-01	-8.660254e-01	1.000000e+00	-8.660254e-
2024-06-30	train	332474.0	1.224647e-16	-2.449294e-16	3.673940e-16	-1.000000e+
2024-07-31	train	437103.0	-5.000000e-01	8.660254e-01	-1.000000e+00	-8.660254e-
2024-08-31	train	417940.0	-8.660254e-01	8.660254e-01	-4.898587e-16	-5.000000e-

direct – sample around the train/validation boundary

	split	arrival_count	cal_Month_sin_1	cal_Month_sin_2	cal_Month_sin_3	cal_Month_cos
date						
2023-09-30	train	352355.0	-1.000000e+00	3.673940e-16	1.000000e+00	-1.836970e-
2023-10-31	train	409924.0	-8.660254e-01	-8.660254e-01	2.388680e-15	5.000000e-
2023-11-30	train	504395.0	-5.000000e-01	-8.660254e-01	-1.000000e+00	8.660254e-
2023-12-31	train	621171.0	-2.449294e-16	-4.898587e-16	-7.347881e-16	1.000000e+
2024-01-31	train	956737.0	5.000000e-01	8.660254e-01	1.000000e+00	8.660254e-
2024-02-29	train	833306.0	8.660254e-01	8.660254e-01	1.224647e-16	5.000000e-
2024-03-31	train	740483.0	1.000000e+00	1.224647e-16	-1.000000e+00	6.123234e-
2024-04-30	train	398587.0	8.660254e-01	-8.660254e-01	-2.449294e-16	-5.000000e-
2024-05-31	train	335652.0	5.000000e-01	-8.660254e-01	1.000000e+00	-8.660254e-
2024-06-30	train	332474.0	1.224647e-16	-2.449294e-16	3.673940e-16	-1.000000e+
2024-07-31	train	437103.0	-5.000000e-01	8.660254e-01	-1.000000e+00	-8.660254e-
2024-08-31	train	417940.0	-8.660254e-01	8.660254e-01	-4.898587e-16	-5.000000e-

```
In [19]: one_year = (
    features['recursive']
    .set_index('date')
    .loc[TRAIN_END - pd.offsets.MonthEnd(11) : TRAIN_END]
    .set_index('cal_Month')
    .sort_index()
)
fig, axes = plt.subplots(1, 2, figsize=(14, 4), sharey=True)
for ax, kind in zip(axes, ['sin', 'cos']):
    for n in range(1, FOURIER_TERMS + 1):
        ax.plot(
            one_year.index,
            one_year[f'cal_Month_{kind}_{n}'],
            marker='o',
            label=f'n = {n}',
        )
    ax.set_title(f'{kind}(2π·n·month / 12)')
    ax.set_xticks(range(1, 13))
```

```
ax.set_xlabel('Month')
ax.legend()
plt.tight_layout()
plt.show()
```



Step 10 — Domain Features (Project-Specific)

These features are **not** in the book. They come from the analysis in notebooks 3–5 and from the calendar. All are known in advance for any future month.

Feature	Definition	Why
<code>is_covid</code>	1 for Jan 2020 – Dec 2021	The structural break. The same regressor as the SARIMAX baseline
<code>is_covid_recovery</code>	1 for Jan – Jun 2022	The rebound period (notebook 5)
<code>is_covid_lag_k</code>	<code>is_covid</code> at each lag of the strategy (book <code>add_lags</code>)	Warns the model that <code>lag_k</code> holds a pandemic value. For example, 2022 rows have a collapsed <code>lag_12</code>
<code>days_in_month</code>	28–31	Monthly totals scale with the number of days

Output. The COVID contamination of `lag_12` in 2021–2022.

```
In [20]: def add_domain_features(df, cfg):
df = df.copy()
df['is_covid'] = df['date'].between(COVID_START, COVID_END).astype(int)
df['is_covid_recovery'] = (
    df['date'].between(POST_COVID_START, COVID_RECOVERY_END).astype(int)
)
df['days_in_month'] = df['date'].dt.days_in_month
added = ['is_covid', 'is_covid_recovery', 'days_in_month']

# flag the lag positions that point into the pandemic (no COVID before 1989)
df, covid_lags = add_lags(df, lags=cfg['lags'], column='is_covid', ts_id=TS_ID)
df[covid_lags] = df[covid_lags].fillna(0).astype(int)
return df, added + covid_lags

for name, cfg in STRATEGIES.items():
    features[name], feature_groups[name]['domain'] = add_domain_features(
```

```

        features[name], cfg
    )

show_step('domain', rows=slice('2021-11-30', '2022-03-31'))

```

	n_features	features
recursive	9	is_covid, is_covid_recovery, days_in_month, is...
direct	7	is_covid, is_covid_recovery, days_in_month, is...

recursive – sample around the train/validation boundary

	split	arrival_count	is_covid	is_covid_recovery	days_in_month	is_covid_lag_1	is_covid_lag_12
date							
2021-11-30	train	107123.0	1	0	30	1	1
2021-12-31	train	174834.0	1	0	31	1	1
2022-01-31	train	326670.0	0	1	31	1	1
2022-02-28	train	311342.0	0	1	28	0	0
2022-03-31	train	276708.0	0	1	31	0	0

direct – sample around the train/validation boundary

	split	arrival_count	is_covid	is_covid_recovery	days_in_month	is_covid_lag_12	is_covid_lag_1
date							
2021-11-30	train	107123.0	1	0	30	1	1
2021-12-31	train	174834.0	1	0	31	1	1
2022-01-31	train	326670.0	0	1	31	1	1
2022-02-28	train	311342.0	0	1	28	1	1
2022-03-31	train	276708.0	0	1	31	1	1

```

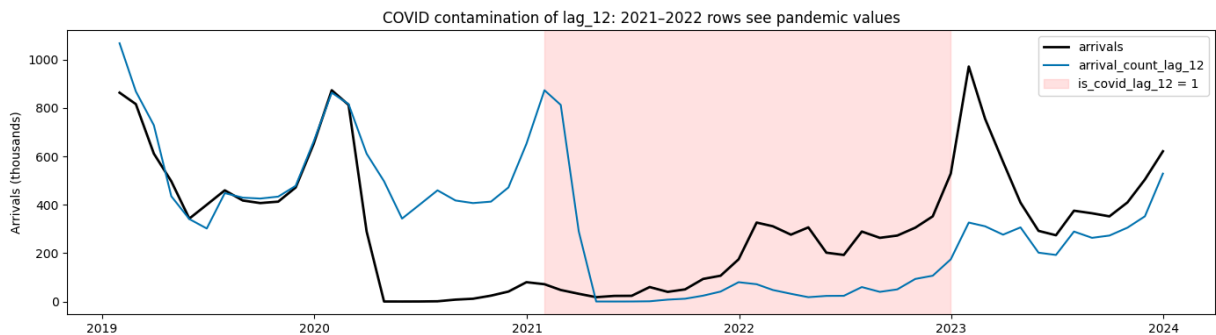
In [21]: frame = features['recursive']
fig, ax = plt.subplots(figsize=(14, 4))
view = frame.query('"2019-01-01" <= date < "2024-01-01"')
ax.plot(view['date'], view[TARGET] / 1e3, color='black', linewidth=2, label='arrival')
ax.plot(view['date'], view[f'{TARGET}_lag_12'] / 1e3, label=f'{TARGET}_lag_12')
contaminated = view.loc[view['is_covid_lag_12'] == 1, 'date']

```

```

ax.axvspan(
    contaminated.min(),
    contaminated.max(),
    color='red',
    alpha=0.1,
    label='is_covid_lag_12 = 1',
)
ax.set_title('COVID contamination of lag_12: 2021-2022 rows see pandemic values')
ax.set_ylabel('Arrivals (thousands)')
ax.legend()
plt.tight_layout()
plt.show()

```



Reading. From January 2021 to December 2022, `lag_12` holds pandemic values that say nothing about the actual level. The same happens to the rolling, seasonal and EWMA features that reach into 2020–2021. `is_covid_lag_k` marks these rows for the model. The further the forecast window moves from the pandemic, the fewer lags reach into it: the table in Step 10 shows which `is_covid_lag_k` flags are still active on the validation rows.

Step 11 — The Feature Pipeline

What. Steps 4–10 are wrapped into one function, `build_feature_matrix(monthly, strategy)`. It applies the same step functions in order and returns the feature table and its feature groups.

- Notebook 7 needs this function to rebuild the recursive features as predictions are fed back.
- Step 12 needs it to rerun the whole pipeline on perturbed data for the leakage check.

Output. The pipeline must reproduce the step-by-step tables exactly.

```

In [22]: FEATURE_STEPS = [
    ('lags', add_lag_features),
    ('rolling', add_rolling_window_features),
    ('seasonal_rolling', add_seasonal_window_features),
    ('ewma', add_ewma_features),
    ('calendar', add_calendar_features),
    ('fourier', add_fourier_month_features),
    ('domain', add_domain_features),
]

```

```
TARGET_DERIVED_GROUPS = ['lags', 'rolling', 'seasonal_rolling', 'ewma']

def build_feature_matrix(monthly_df, strategy):
    """Apply every feature step to a copy of the monthly series for one strategy."""
    cfg = STRATEGIES[strategy]
    df = monthly_df.copy()
    groups = {}
    for group, step in FEATURE_STEPS:
        df, groups[group] = step(df, cfg)
    return df, groups

for name in STRATEGIES:
    rebuilt, groups = build_feature_matrix(monthly, name)
    pd.testing.assert_frame_equal(rebuilt, features[name])
    assert groups == feature_groups[name]
    print(
        f'{name:<9}: pipeline reproduces the step-by-step table '
        f'({rebuilt.shape[1]} columns)'
    )

display(
    pd.DataFrame({
        name: {group: len(cols) for group, cols in groups.items()}
        for name, groups in feature_groups.items()
    }).assign(**{'target-derived': lambda d: d.index.isin(TARGET_DERIVED_GROUPS)})
)
```

```
recursive: pipeline reproduces the step-by-step table (49 columns)
direct    : pipeline reproduces the step-by-step table (45 columns)
```

	recursive	direct	target-derived
lags	6	4	True
rolling	12	12	True
seasonal_rolling	4	4	True
ewma	3	3	True
calendar	5	5	False
fourier	6	6	False
domain	9	7	False

Step 12 — Leakage Check

What. A feature leaks if the value at row `t` depends on data that would not be available when forecasting `t`. The test checks this directly instead of trusting each `shift`:

1. Pick a cutoff date and **multiply every value after it by 10**.
2. Rebuild the features with the pipeline.

3. Rows up to `cutoff + min_shift` must keep **identical** target-derived features, since they may only see data up to the cutoff.
4. Negative control: the next row must **change**, which proves the test can detect a difference. It does not apply when that row lies beyond the data (`None`).

For the direct strategy, `cutoff = TRAIN_END` covers the whole validation window. None of the validation features may depend on validation values.

```
In [23]: def target_derived_columns(groups):
    return [col for group in TARGET_DERIVED_GROUPS for col in groups[group]]

def check_no_leakage(strategy, cutoff):
    min_shift = STRATEGIES[strategy]['min_shift']
    base, groups = build_feature_matrix(monthly, strategy)
    perturbed = monthly.copy()
    perturbed.loc[perturbed['date'] > cutoff, TARGET] *= 10
    shocked, _ = build_feature_matrix(perturbed, strategy)

    cols = target_derived_columns(groups)
    last_safe = cutoff + pd.offsets.MonthEnd(min_shift)
    safe = base['date'] <= last_safe
    first_unsafe = base['date'] == last_safe + pd.offsets.MonthEnd(1)
    unchanged = np.allclose(
        base.loc[safe, cols], shocked.loc[safe, cols], equal_nan=True
    )
    # control is not applicable when the next row is beyond the data
    detects = (
        not np.allclose(
            base.loc[first_unsafe, cols],
            shocked.loc[first_unsafe, cols],
            equal_nan=True,
        )
        if first_unsafe.any()
        else None
    )
    return {
        'strategy': strategy,
        'cutoff': f'{cutoff:%b %Y}',
        'rows that must not change': f'up to {last_safe:%b %Y}',
        'unchanged (no leakage)': unchanged,
        'next row changes (control)': detects,
    }

leakage = pd.DataFrame([
    check_no_leakage(strategy, cutoff)
    for strategy in STRATEGIES
    for cutoff in [pd.Timestamp('2015-06-30'), TRAIN_END]
])
display(leakage)
assert leakage['unchanged (no leakage)'].all()
assert leakage['next row changes (control)'].dropna().all()
```

	strategy	cutoff	rows that must not change	unchanged (no leakage)	next row changes (control)
0	recursive	Jun 2015	up to Jul 2015	True	True
1	recursive	Aug 2024	up to Sep 2024	True	True
2	direct	Jun 2015	up to Jun 2016	True	True
3	direct	Aug 2024	up to Aug 2025	True	None

Step 13 — Warm-up Rows and Test-Row Masking

What. Two cleaning rules turn the working tables into model-ready data.

1. **Warm-up.** The first rows cannot have all of their features: `lag_36` needs 36 months of history, and the 3-year seasonal window also reaches back 36 months. These rows are dropped. The loss is small, since the series starts in 1989.
2. **Recursive validation rows.** In the recursive table, the target-derived features of the validation rows were computed from the **actual** validation values (Step 4). They are set to `NaN`, so the saved file cannot be used for an accidental one-step-ahead evaluation. Notebook 7 fills them month by month from the model's predictions, using `build_feature_matrix`.

The direct table needs no masking: Step 12 proved its validation features only use training data.

```
In [24]: model_ready = {}
for name, frame in features.items():
    cols = target_derived_columns(feature_groups[name])
    complete = frame[cols].notna().all(axis=1)
    first_complete = frame.loc[complete, 'date'].min()
    ready = frame[frame['date'] >= first_complete].reset_index(drop=True)
    train_nans = int(ready.loc[ready['split'] == 'train', cols].isna().sum().sum())
    if name == 'recursive':
        ready.loc[ready['split'] == 'val', cols] = np.nan
    model_ready[name] = ready
    print(
        f'{name:<9}: first complete row {first_complete:%b %Y} | '
        f'dropped {len(frame) - len(ready)} warm-up rows | '
        f'{ready.query("split == \'train\'").shape[0]} train + '
        f'{ready.query("split == \'val\'").shape[0]} validation rows | '
        f'NaNs left in train: {train_nans}'
    )
```

recursive: first complete row Jan 1992 | dropped 36 warm-up rows | 392 train + 12 validation rows | NaNs left in train: 0
direct : first complete row Jan 1992 | dropped 36 warm-up rows | 392 train + 12 validation rows | NaNs left in train: 0

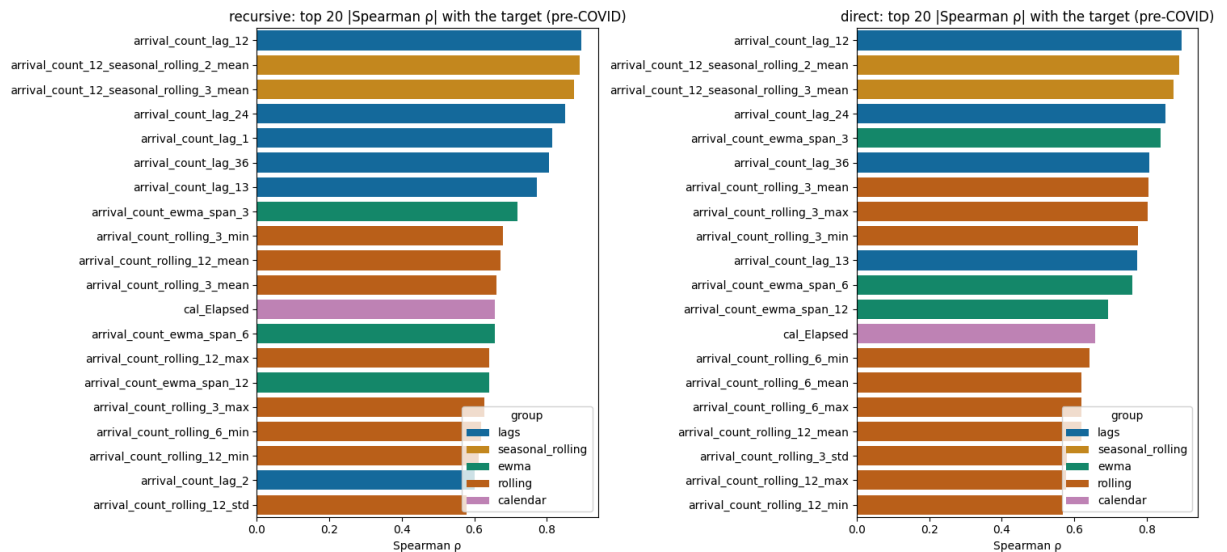
Step 14 — Feature Overview

What. A first look at how each feature relates to the target. This is **not** feature selection (planned for notebook 7), only a sanity check that the features carry signal.

How. The Spearman (rank) correlation with the target on the **pre-COVID training rows**, since the pandemic would dominate any correlation. `split`, the id and the target are excluded. The top features of each strategy are shown.

```
In [25]: def feature_columns(groups):
          return [col for cols in groups.values() for col in cols]

TOP_N = 20
fig, axes = plt.subplots(1, 2, figsize=(15, 7))
for ax, (name, frame) in zip(axes, model_ready.items()):
    pre = frame.query('split == "train" and date < @COVID_START')
    cols = [
        c
        for c in feature_columns(feature_groups[name])
        if not (pre[c] == pre[c].iloc[0]).all()
    ]
    corr = pre[cols].corrwith(pre[TARGET], method='spearman')
    top = corr.reindex(corr.abs().sort_values(ascending=False).index).head(TOP_N)
    group_of = {c: g for g, cs in feature_groups[name].items() for c in cs}
    sns.barplot(
        x=top.values, y=top.index, hue=top.index.map(group_of), dodge=False, ax=ax
    )
    ax.set_title(f'{name}: top {TOP_N} |Spearman p| with the target (pre-COVID)')
    ax.set_xlabel('Spearman p')
    ax.set_ylabel('')
    ax.legend(title='group', loc='lower right')
plt.tight_layout()
plt.show()
```



Step 15 — Save the Feature Tables

Outputs (in `data/processed/`), consumed by notebook 7:

File	Content
<code>features_recursive.parquet</code>	Recursive table: train rows complete, validation target-derived features masked (<code>NaN</code>)
<code>features_direct.parquet</code>	Direct table: complete for train and validation
<code>feature_groups.json</code>	Target and id columns, the feature columns of each strategy by group, and the parameters used

Every table keeps `date`, `series_id`, `split` and the **raw** target `arrival_count`. The target transformation (`none` / `log` / `AutoStationaryTransformer`) is applied in notebook 7 and inverse-transformed before scoring, as in notebook 5.

```
In [26]: feature_spec = {
    'target': TARGET,
    'id_columns': ['date', TS_ID, 'split'],
    'horizon': HORIZON,
    'target_derived_groups': TARGET_DERIVED_GROUPS,
    'strategies': {
        name: {
            **STRATEGIES[name],
            'file': f'features_{name}.parquet',
            'groups': feature_groups[name],
        }
        for name in STRATEGIES
    },
    'parameters': {
        'rolling_windows': ROLLING_WINDOWS,
        'rolling_aggs': ROLLING_AGGS,
        'seasonal_period': SEASONAL_PERIOD,
        'seasonal_windows': SEASONAL_WINDOWS,
```

```

        'seasonal_aggs': SEASONAL_AGGs,
        'ewma_spans': EWMA_SPANS,
        'fourier_terms': FOURIER_TERMS,
    },
}

for name, frame in model_ready.items():
    path = os.path.join(DATA_FOLDER_PATH, f'features_{name}.parquet')
    frame.to_parquet(path, index=False)
    print(f'{path} → {frame.shape}')

spec_path = os.path.join(DATA_FOLDER_PATH, 'feature_groups.json')
with open(spec_path, 'w', encoding='utf-8') as f:
    json.dump(feature_spec, f, indent=2)
print(f'{spec_path} written')

```

```

c:\git_reps\brazil_tourism_forecasting_benchmark\data\processed\features_recursive.p
arquet → (404, 49)
c:\git_reps\brazil_tourism_forecasting_benchmark\data\processed\features_direct.parq
uet → (404, 45)
c:\git_reps\brazil_tourism_forecasting_benchmark\data\processed\feature_groups.json
written

```

Key Takeaways

What was built. A reusable pipeline (`build_feature_matrix`) that turns the monthly series into two model-ready tables, from the first complete month (after 36 warm-up months) to the last validation month (the test set is not included). Step 13 prints the exact date range and row counts:

Strategy	Features	Target-derived	Temporal + domain
recursive	45	25 (lags 1, 2, 12, 13, 24, 36; rolling; seasonal; EWMA, shift ≥ 1)	20
direct	41	23 (lags 12, 13, 24, 36; rolling; seasonal; EWMA, shift ≥ 12)	18